

Advanced (Habanero)

Q1. What is the vertex of the parabola $y = x^2 + 4x + 4$?

- (A) $(-2, 0)$
- (B) $(-2, 4)$
- (C) $(0, 4)$
- (D) $(2, 0)$
- (E) $(0, 0)$

Q2. What is the axis of symmetry of the parabola $y = x^2 - 6x + 8$?

- (A) $x = -2$
- (B) $x = 2$
- (C) $x = 3$
- (D) $x = 5$
- (E) $x = 0$

Q3. What are the x -intercepts of the parabola $y = x^2 - 4$?

- (A) $(-2, 0)$ and $(2, 0)$
- (B) $(-1, 0)$ and $(1, 0)$
- (C) $(-3, 0)$ and $(3, 0)$
- (D) $(-4, 0)$ and $(4, 0)$
- (E) $(-5, 0)$ and $(5, 0)$

Q4. What is the direction of the parabola $y = -x^2 + 2$?

- (A) Up
- (B) Down
- (C) Left
- (D) Right
- (E) None

Q5. What is the maximum value of the parabola $y = -x^2 + 4x - 3$?

- (A) -3
- (B) 1
- (C) 2
- (D) 4
- (E) 5

Q6. Which parabola has the equation $y = (x - 2)^2 - 1$?

- (A) Vertex at $(2, -1)$
- (B) Vertex at $(-2, 1)$
- (C) Vertex at $(0, 0)$
- (D) Vertex at $(1, 1)$
- (E) Vertex at $(0, -1)$

Q7. What is the y -intercept of the parabola $y = x^2 - 2x - 3$?

- (A) -3
- (B) -2
- (C) -1
- (D) 0
- (E) 1

Q8. Which of the following is a characteristic of the parabola $y = x^2 + 2x + 1$?

- (A) Axis of symmetry at $x = -1$
- (B) Axis of symmetry at $x = 1$
- (C) Vertex at $(0, 0)$
- (D) Vertex at $(1, 0)$
- (E) Vertex at $(-1, 0)$

Open Ended Questions (FRQ)**Q9.** Graph the parabola $y = x^2 - 4x - 3$ and identify its vertex, axis of symmetry, and x -intercepts.**Q10.** Find the maximum value of the parabola $y = -x^2 + 2x + 1$ and show your work.**Chef's Tips**

- Tip 1: Make sure to check for any calculation errors.
- Tip 2: Use a graphing calculator to verify your answers.
- Tip 3: Read each question carefully and follow the instructions.